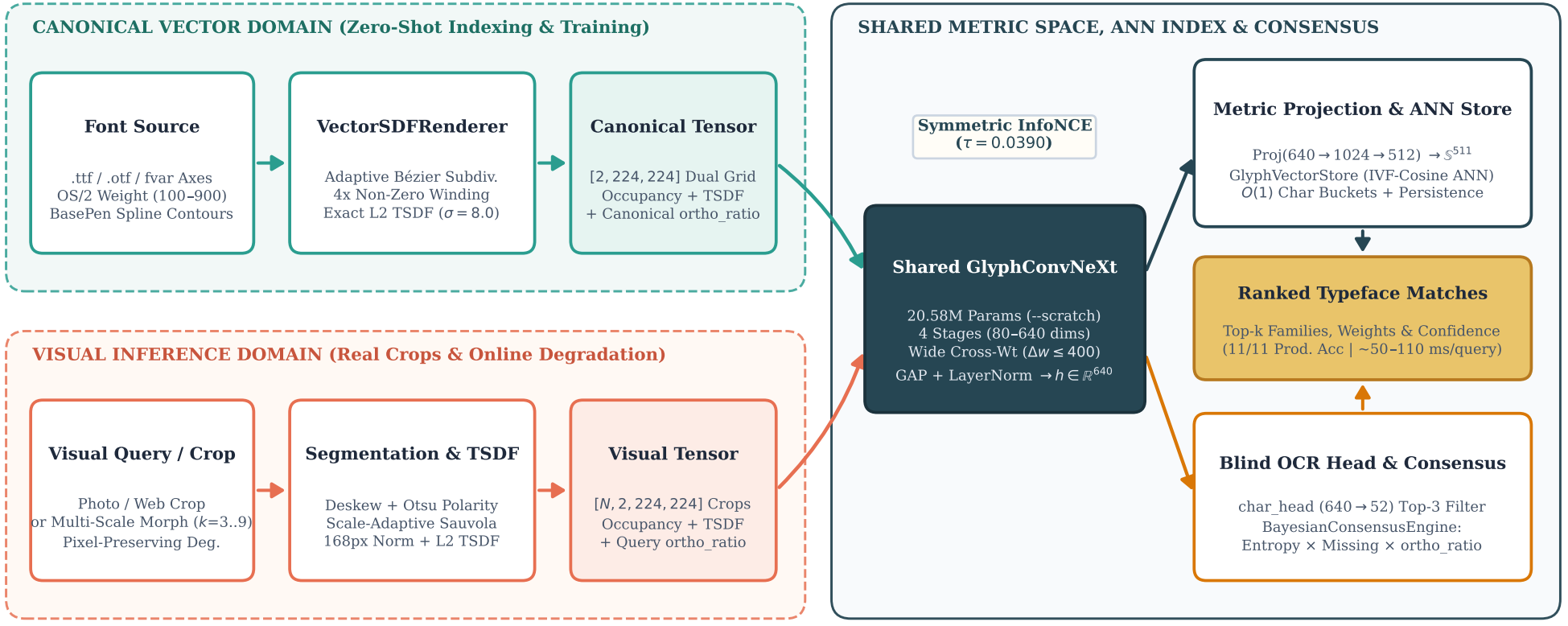

CANONICAL VECTOR DOMAIN (Zero-Shot Indexing & Training)
Font Source
.ttf / .otf / fvar Axes
OS/2 Weight (100–900)
BasePen Spline Contours
VectorSDFRenderer
Adaptive Bézier Subdiv.
4x Non-Zero Winding
Exact L2 TSDF (σ = 8.0)
Canonical Tensor
[2, 224, 224] Dual Grid
Occupancy + TSDF
+ Canonical ortho_ratio
VISUAL INFERENCE DOMAIN (Real Crops & Online Degradation)
Visual Query / Crop
Photo / Web Crop
or Multi-Scale Morph (k=3..9)
Pixel-Preserving Deg.
Segmentation & TSDF
Deskew + Otsu Polarity
Scale-Adaptive Sauvola
168px Norm + L2 TSDF
Visual Tensor
[N, 2, 224, 224] Crops
Occupancy + TSDF
+ Query ortho_ratio
SHARED METRIC SPACE, ANN INDEX & CONSENSUS
Symmetric InfoNCE
(τ = 0.0390)
Shared GlyphConvNeXt
20.58M Params (--scratch)
4 Stages (80–640 dims)
Wide Cross-Wt (Δw ≤ 400)
GAP + LayerNorm → h ∈ ℝ^640
Metric Projection & ANN Store
Proj(640 → 1024 → 512) → 𝕊^511
GlyphVectorStore (IVF-Cosine ANN)
O(1) Char Buckets + Persistence
Ranked Typeface Matches
Top-k Families, Weights & Confidence
(11/11 Prod. Acc | ~50–110 ms/query)
Blind OCR Head & Consensus
char_head (640 → 52) Top-3 Filter
BayesianConsensusEngine:
Entropy × Missing × ortho_ratio